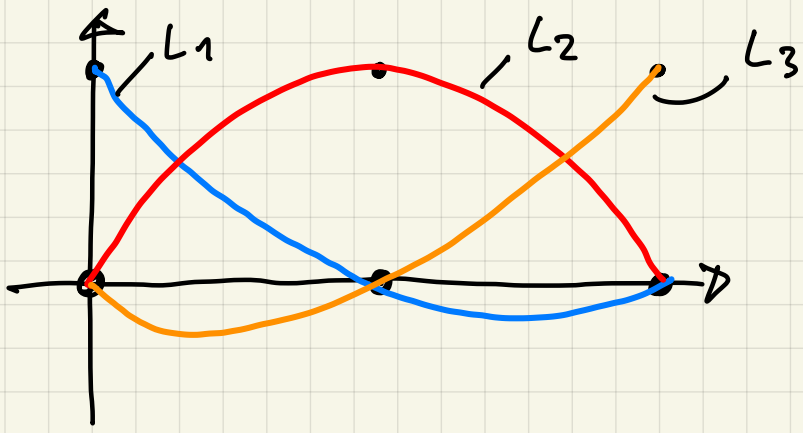
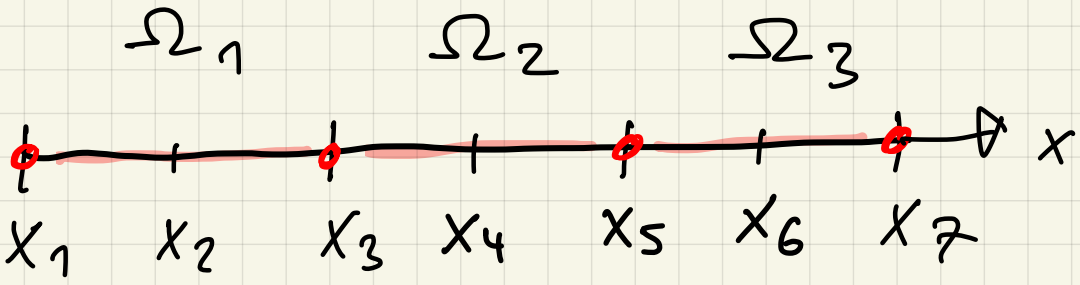
Abschnittsweise quadratische Basisfunktionen

Lagrange Polynome

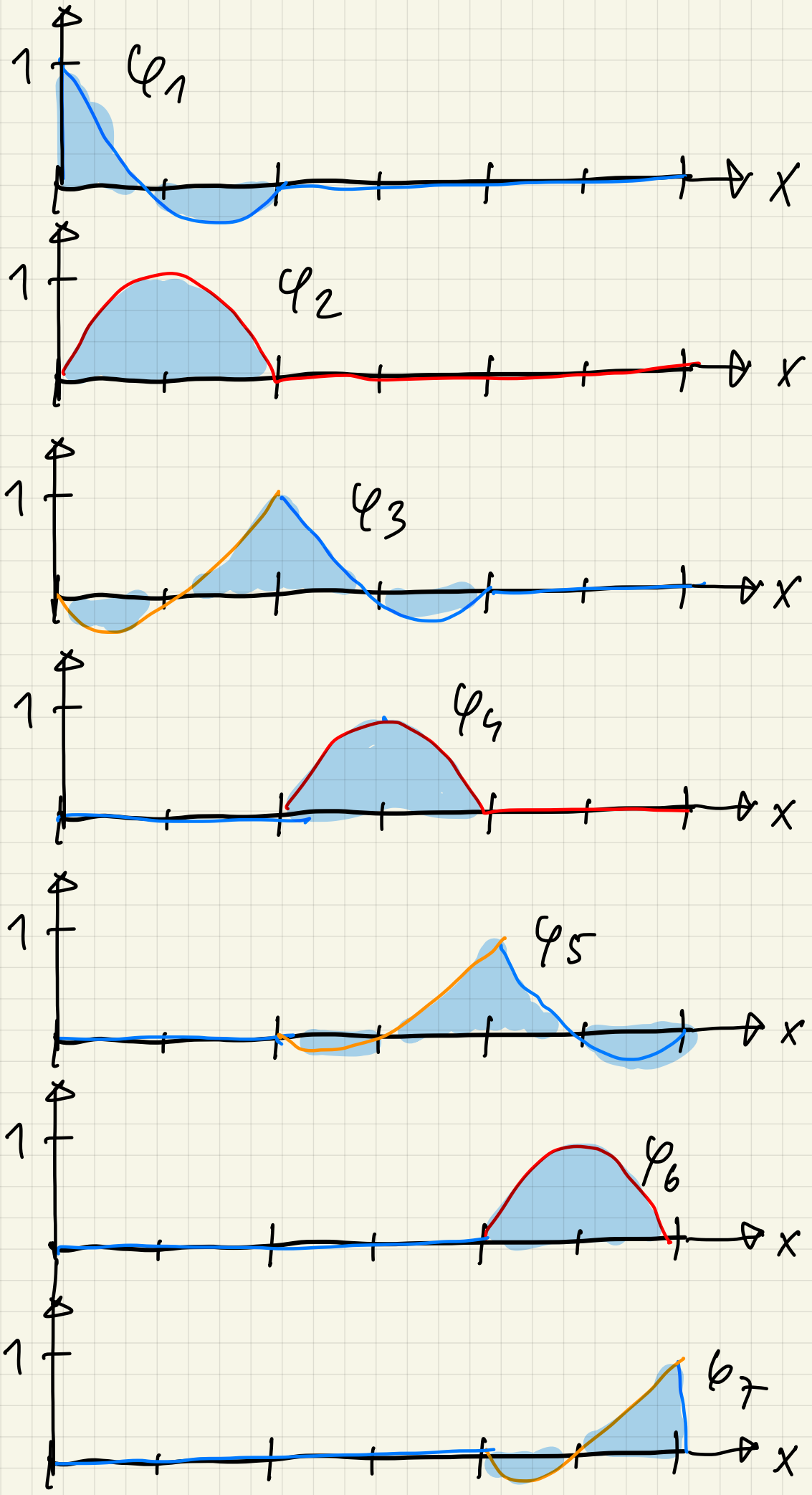
$p=2$



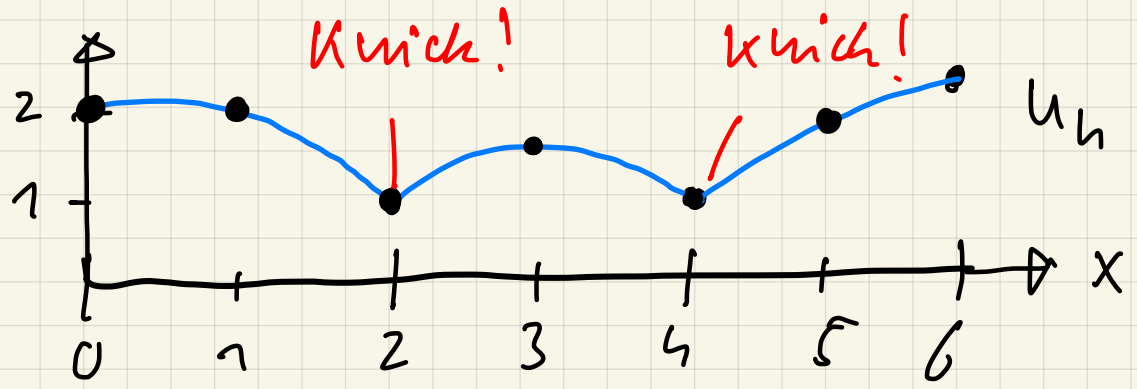
Elemente und Knoten



Abschnittsweise quadratische Basisfunktionen



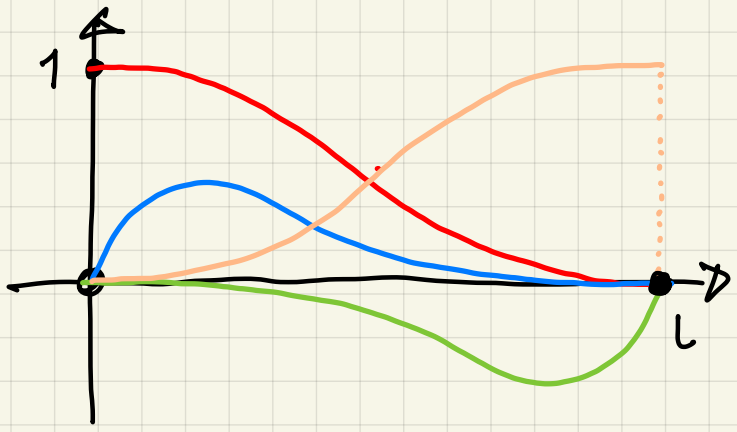
2.



3. Die Funktion u_h ist an den Knoten zwischen zwei Elementen nicht differenzierbar, sonst schon.

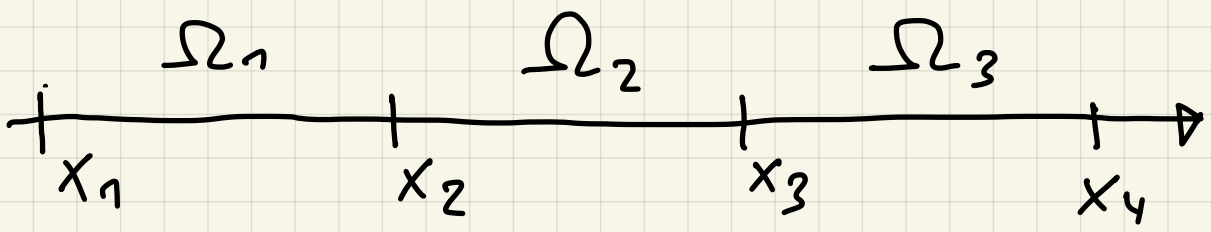
Basisfunktionen mit stetigen Ableitungen

Kubische Hermite Polynome

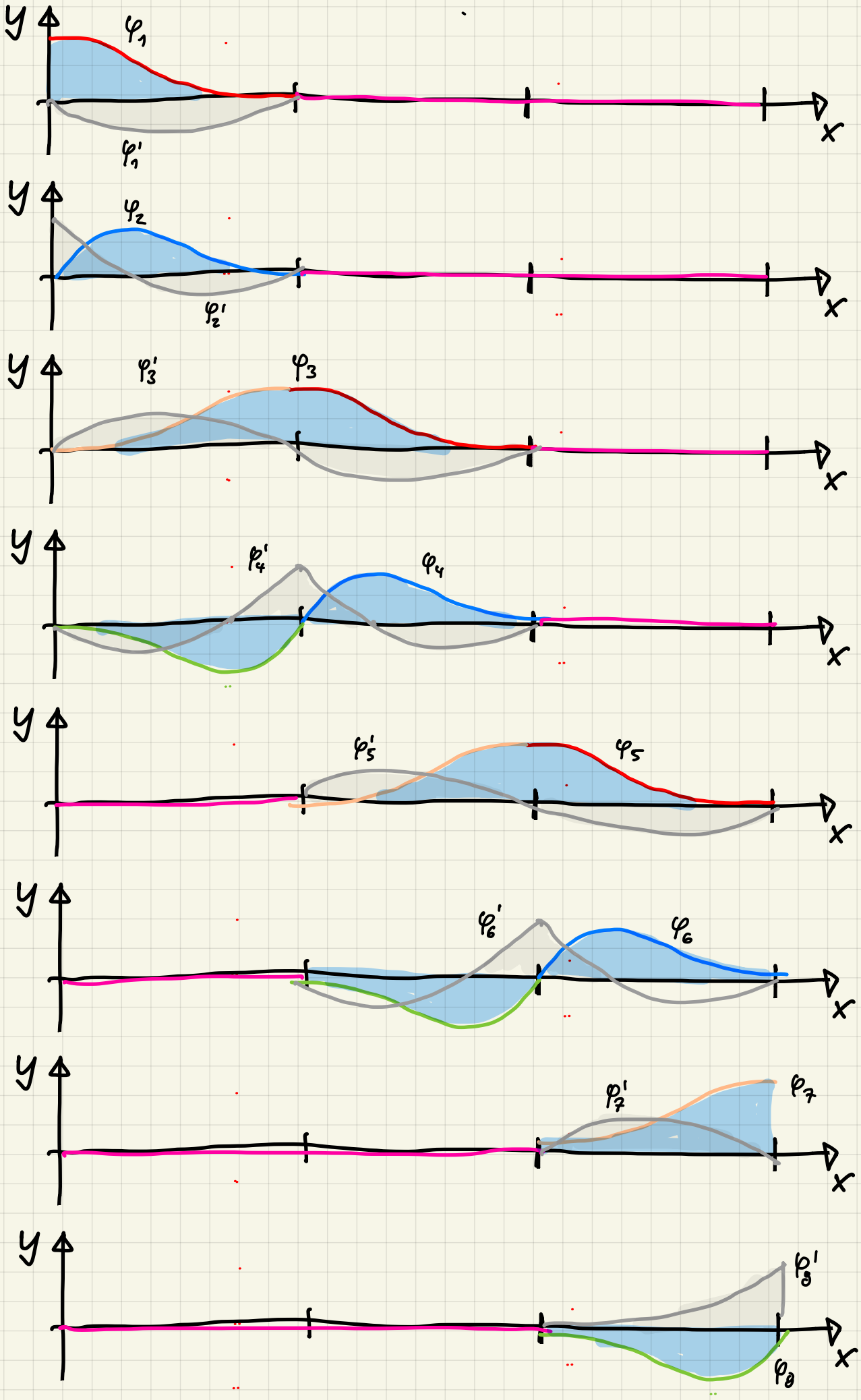


See Fish & Belytschko p. 259

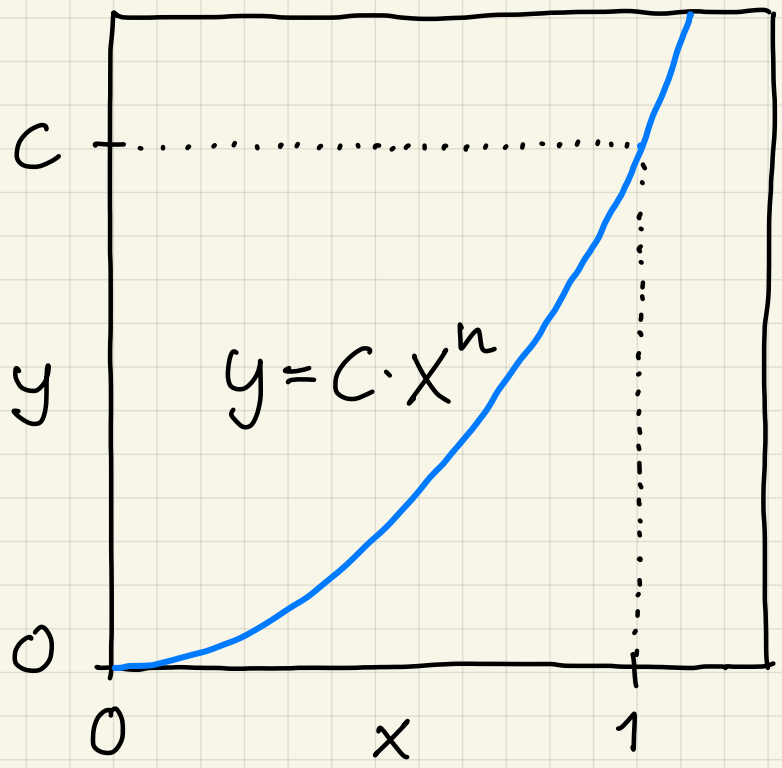
Elemente und Knoten



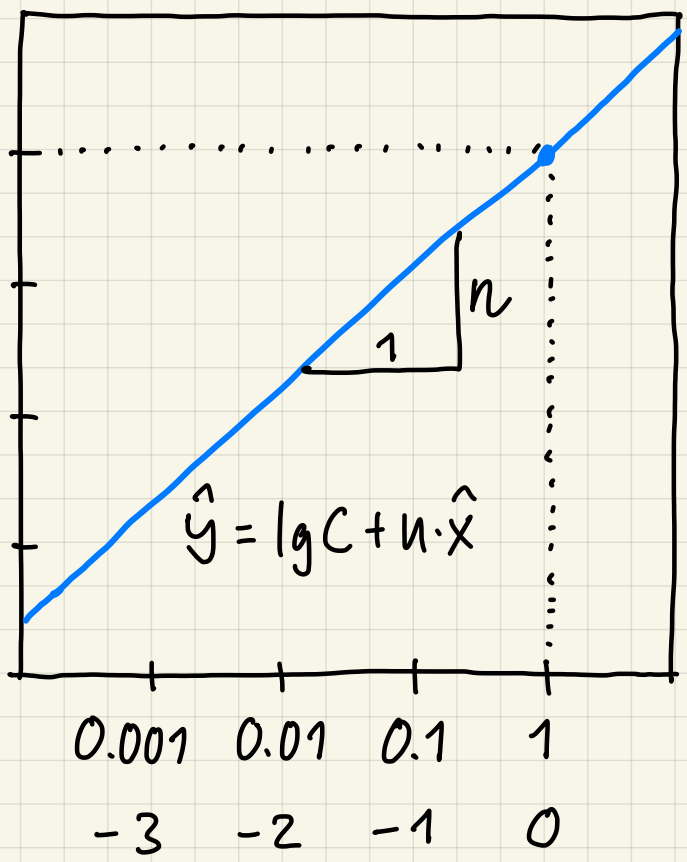
Abschnittsweise kubische Basisfunktionen mit stetiger Ableitung



Log-Log-Plot



$\hat{y}$	y
$0 \cdot \lg c$	$1 \cdot c$
$-1 \cdot \lg c$	$0.1 \cdot c$
$-2 \cdot \lg c$	$0.01 \cdot c$
$-3 \cdot \lg c$	$0.001 \cdot c$



Beziehungen zwischen Koordinaten

$y = 10^{\hat{y}} \Leftrightarrow \hat{y} = \lg y$

$x = 10^{\hat{x}} \Leftrightarrow \hat{x} = \lg x$

Gleichung in x, y

$$y = c \cdot x^n$$

Gleichung in $\hat{x}, \hat{y}$

$$\hat{y} = \lg y = \lg(c \cdot x^n) = \lg c + \lg x^n$$

$$= \lg c + \lg (10^{\hat{x}})^n = \lg c + \lg 10^{\hat{x} \cdot n}$$

$$\hat{y} = \lg c + n \cdot \hat{x}$$